

PAPER<i>343 ■ SGR J1745-2900: SC</i>m Mass-Modified Luminosity and Doubled f_{react} (June 2013)

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Session: 96

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF treatment of magnetar SCm mass modification with $L_X = \rho_{vac} f_{res} V$ **Author:** Daniel T. Murphy

Abstract

A novel UQFF form for the superconductive modifier SCm of SGR J1745-2900 is derived as a mass-dependent suppression by the critical field ratio: $SC_m = M(1 - B/B_{crit})$. The X-ray luminosity is expressed as $L_X = \rho_{vac} f_{res} V$, coupling vacuum energy density, resonance frequency, and magnetospheric volume. The activation event of June 2013 corresponds to a doubling of f_{react} , confirmed by the sudden spin-up and luminosity jump. $T_{surf} = 1.16 \times 10^7$ K is derived from the Stefan-Boltzmann radiative balance.

2. Core Physics

2.1 Mass-Modified Superconductive Modifier

$$SC_m = M \cdot \left(1 - \frac{B}{B_{crit}}\right)$$

where $B_{crit} = 4.4 \times 10^{11}$ T (quantum critical field). For SGR J1745-2900: $B = 2 \times 10^{11}$ T $\ll B_{crit}$, giving $SC_m \approx M$ (nearly full superconductive coupling).

2.2 Vacuum-Energy X-ray Luminosity Form

$$L_X = \rho_{vac} \cdot f_{res} \cdot V_{mag}$$

where $\rho_{vac} = \rho_{SCm} - \rho_{UA}$ is the net vacuum energy density and V_{mag} is the magnetospheric volume.

2.3 June 2013 Activation Event

$$f_{react}^{post} = 2 \cdot f_{react}^{pre}$$

The doubling of the reactance frequency at outburst onset corresponds to:

$$\Delta L_X = \rho_{vac} \cdot \Delta f_{res} \cdot V_{mag} = \rho_{vac} \cdot f_{react}^{pre} \cdot V_{mag}$$

2.4 Surface Temperature

From Stefan-Boltzmann balance of magnetospheric luminosity:

$$T_{\rm surf} = \left(\frac{L_X}{4\pi R_{\rm NS}^2 \sigma_{\rm SB}}\right)^{1/4} = 1.16 \times 10^7 \, \mathrm{K}$$

3. Key Values

Quantity	Formula	Value
SC_m	$M(1-B/B_{\rm crit})$	$\blacksquare M_{\rm NS} (B \blacksquare B_{\rm crit})$
L_X	$?vac\blacksquare f_{\rm res}\blacksquare V$	$\sim 10\blacksquare 5 \text{ erg/s}$
f_react (pre-2013)	canonical	f0
f_react (post June 2013)	2f0	2f0
T_surf	Stefan-Boltzmann	$1.16\blacksquare 10^7 \text{ K}$
B_crit	Quantum critical	$4.4\blacksquare 10\blacksquare\blacksquare \text{ T}$

4. Physical Significance

SGR J1745-2900 is the only magnetar within 0.3 pc of Sgr A*, making it the unique test-bed for UQFF near the Galactic Center supermassive black hole. The $SC_m = M(1-B/B_{\rm crit})$ form establishes that even strongly magnetized neutron stars maintain near-unity superconductive coupling due to $B \blacksquare B_{\rm crit}$. The $L_X = ?vac\blacksquare f_{\rm res}\blacksquare V$ form is a direct observable prediction: doubling $f_{\rm res}$ (as seen in June 2013) should produce a factor of 2 luminosity jump, consistent with the 2013 XMM-Newton observations.

5. Deduplication Note

****vs. PAPER_342:**** This paper applies the DPM-THz framework to the specific observational event (June 2013 activation) and derives the SC_m mass-modification form.

vs. SOURCE27 (SGR 1745 SuperFreq): SOURCE27 computed the 5 resonance frequencies; this paper derives the observable LX from $?vac\blacksquare f_{\rm res}\blacksquare V$.

6. Classification

Physics Territory: FIRST UQFF SC_m mass-modified form for magnetar near Galactic Center **Scale:** Stellar/Galactic Center (0.3 pc) **CP Implementation:** SGR17452900SCmLxFreqFormCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq]\blacksquare \exp(-? \blacksquare t) = 1 - 5.7\text{e-}1 \blacksquare \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_U at event horizon = $2.0\text{e+}18 \text{ m/s}\blacksquare$.